Age and Tectonic Implications of Paleomagnetic Results From Plutons Near Prince Rupert, British Columbia

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A total of 25 sites (125 cores, 250 specimens) were selected from four plutons along the Skeena River cross-section of the Coast plutonic complex. After af demagnetization a stable remanence direction was isolated for the diorite of the Kasiks pluton, the gneissic diorite and quartz diorite of the Quottoon pluton, and the diorite and quartz diorite of the Ecstall pluton but not for the granodiorite phase of the Ponder pluton. The Kasiks-Quottoon pole (144°W, 76°N ($P_{0.05}$: 7°, 7°)) is concordant with other Eocene poles from cratonic North America, and thus no postemplacement tectonic rotation or translation is indicated for these plutons. The Ecstall pole (47°W, 68°N ($P_{0.05}$: 13°, 16°)) in Greenland is discordant with other Upper Cretaceous poles. This pluton has probably undergone a westerly tilt of ≃20°. The Kasiks and Quottoon plutons most likely cooled quickly within the time span of one normal polarity interval (<½ m.y.) and the Ecstall pluton within the time span of one normal and one reversed interval (<1 m.y.). This result supports the hypothesis that the pole positions and K-Ar ages of 43 ± 5, 48 ± 7, and 75 ± 12 m.y. for the plutons date the age of their emplacement rather than the age of their unroofing or of their thermal metamorphism, which would give different paleomagnetic results. Acceptance of the K-Ar dates as the age of emplacement suggests that the Coast plutonic complex evolved at the western margin of the North American plate as three prolonged episodes of plutonic emplacement to form the western (84–140 m.y.), central (64–80 m.y.), and eastern (40–50 m.y.) K-Ar date zones.

The age of emplacement of the plutonic rocks that form much of the NW-trending core of the Coast Mountains (Figure 1) is an unresolved problem in tracing the evolution of the Coast plutonic complex [*Hutchison*, 1967]. Geologic field relations allow any age between Late Jurassic and late Miocene. The K-Ar ages on biotite and hornblende separates from the plutons form three distinct age zones that parallel the trend of the Coast Mountains [*Hutchison*, 1970; *Monger and Hutchison*, 1971]. The eastern zone yields K-Ar ages of 40–50 m.y., the central zone ages of 64–80 m.y., and the western zone ages of 84–140 m.y. *Hutchison* [1967] has pointed out that these dates may represent the actual time of emplacement, or the time of regional thermal metamorphic events, or the time at which radiogenic argon ceased to diffuse during gradual uplift and unroofing. Because the paleomagnetic method for determining age from the pole position of a rock unit is less sensitive to thermal updating than the K-Ar method, this study was undertaken as an initial test. The 25 sites (Figure 2) were spread along an 80-km cross-section of the complex to minimize the chance of tectonic rotations, which could move the pole positions and thereby negate the paleomagnetic method. The section follows the Skeena River eastward from Prince Rupert, British Columbia (Figure 2).

GEOLOGY

The sampled section is in the southeastern corner of the Prince Rupert–Skeena map area [*Hutchison*, 1967], which spans the Coast plutonic complex. The oldest rock units in the map area are sediments and volcanics that have been metamorphosed from lower-grade greenschist facies on the eastern side of the complex to high-grade gneisses and migmatites of the amphibolite facies in the central portion and on the western side [*Roddick and Hutchison*, 1972]. On the basis of correlations with unmetamorphosed stratigraphic equivalents outside the map area these units are thought to be Carboniferous, Triassic, and Jurassic. On the extreme eastern side of the complex the Ponder pluton has intruded unmetamorphosed argillites and graywackes of the Bowser group. The Bowser group contains Late Jurassic and possibly earliest Lower Cretaceous fossils in adjacent map areas [*Duffell and Souther*, 1964; *Eisbacher*, 1973]. This then represents a maximum age for emplacement of the plutons. Additional evidence for intrusion commencing in the Lower Cretaceous comes from the conglomerate sediments flanking the complex. Granitic clasts are rare in Jurassic and older sediments and could have been derived from lower Paleozoic plutons in southeastern Alaska [*Brew et al.*, 1966]. Conversely, abundant granitic clasts occur in Lower Cretaceous and younger sediments and reflect emplacement and erosion of the plutons.

The individual plutons within the complex are elongate and parallel the NW-SE trend of the Coast Mountains. They are up to 80 or more kilometers long and up to 30 or so kilometers wide. They commonly grade in composition from basic or intermediate at their perimeters to acidic in the cores (Figure 2). The perimeter phase is most commonly diorite, which is often well foliated or gneissic and may have associated gabbroic zones. Progressing inward, the plutons grade to quartz diorite and then to granodiorite and finally to quartz monzonite in the core. During this gradation the rocks become increasingly massive or less foliated, and hornblende gives way to biotite as the dominant mafic mineral. The petrography and structure of these plutons have been summarized by *Hutchison* [1970].

Of the four plutons sampled in this study, three come from the eastern age zone. These are the Ponder pluton, which gives K-Ar ages on biotite of 46 ± 10 and 47 ± 4 m.y. (G.S.C. 65-32 and 66-15 from *Wanless et al.* [1967, 1968]), the Kasiks pluton, which gives a K-Ar age on biotite of 43 ± 5 m.y. (G.S.C. 65-29), and the Quottoon pluton, which gives K-Ar ages on biotite and hornblende pairs of 50 ± 5 and 48 ± 9 m.y. (G.S.C. 66-6 and 66-7) and 44 ± 5 and 49 ± 7 m.y. (G.S.C. 66-9 and 66-8). The fourth pluton is the Ecstall

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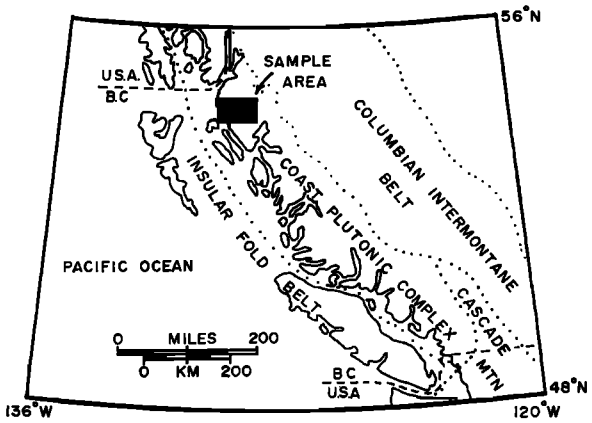


Fig. 1. Location map.

pluton, which comes from the central age zone. It gives K-Ar ages of 64 ± 8 m.y. (G.S.C. 65-31) on biotite and of 70 ± 4 and 87 ± 15 m.y. (G.S.C. 66-13 and 66-12) on a biotite and hornblende pair. The age of the Ecstall pluton is set at 75 ± 12 m.y., and although it is not closely defined, acceptance of a somewhat greater age would not materially affect the paleomagnetic conclusions that follow from comparison with other Upper Cretaceous data.

The minimum age for emplacement of the plutons is given by the K-Ar dates. The minimum age from field relations comes from unmetamorphosed and undeformed basic dikes that cut the plutons along Portland Inlet to the north [*Hutchison*, 1967] and in Laredo Sound and Bella Coola map areas to the south [*Baer*, 1967]. These dikes give K-Ar ages of 12.5 and 14.5 m.y. (G.S.C. 67-32 and 67-31 from *Wanless et al.* [1970]) and a paleomagnetic pole position that is concordant with other poles from North American Miocene rock units [*Symons*, 1969]. These dikes are thought to be genetically related to localized late Tertiary volcanic centers such as the volcanic neck found just north of the Ecstall pluton and the flat-lying rhyolitic to basaltic flows found in Laredo Sound, Bella Coola, and Hecate Strait map areas [*Roddick*, 1970].

SAMPLING

As is shown in Figure 2, the 25 sampling sites are spread out fairly uniformly along the 80-km section cut by the Skeena River, if the nonplutonic rocks are excluded. The specimens of sites 1 and 2 are from a massive zone, and those of sites 3 and 4 are from a foliated zone in the granodiorite phase of the Ponder pluton. Sites 5–8 represent the diorite phase of the Kasiks pluton. Sites 9–12 are from the foliated quartz diorite phase of the Quottoon pluton. Sites 13–16 are from a gneissic diorite phase that forms the boundary phase of either the Quottoon or the Ecstall pluton. Finally, sites 17–20 and 21–25 are from the massive diorite and quartz diorite phases, respectively, of the Ecstall pluton.

EXPERIMENTAL PROCEDURE

Five cores were collected at each of the 25 sites (Figure 2). The cores were oriented in situ by means of a solar compass. Brunton compass and topographic sightings were used for checks. Two specimens were cut from each core. The natural remanent magnetization (NRM) of each specimen was measured on an automated biastatic magnetometer [*Larochelle and Christie*, 1967]. One pilot specimen per site with an average NRM direction and intensity was step-demagnetized [*Larochelle and Black*, 1965] in alternating field (af) peak intensities of 50, 100, 150, 200, 300, 400, 600, and 800 Oe, its remanence being remeasured after each step. By using the stability index values calculated from the remanence directions [*Tarling and Symons*, 1967] and the intensity decay curves as a guide an af intensity was selected (Table 1) in which the remaining site specimens were demagnetized before their characteristic remanent magnetizations (RM) were remeasured. For 10 sites the specimens were af-demagnetized at a second higher field intensity and remeasured. The results from the cleaning treatment giving the best clustering of characteristic RM directions were used for statistical analysis.

The af step demagnetization of the pilot specimens indicated two types of remanence response. The first type was found for 11 sites (sites 10, 15–24) and indicated a highly stable RM. For these the median destructive field (i.e., the af

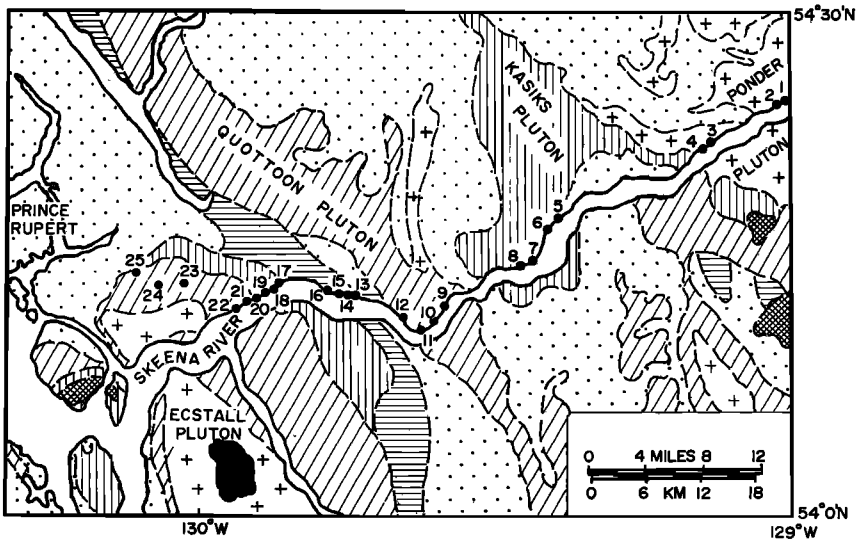


Fig. 2. Sampling sites and generalized geology after *Hutchison* [1967]. The petrologic phases of the plutons distinguished are gabbro (cross-hatching), gneissic diorite (horizontal lines), diorite (vertical lines), quartz diorite (sloped lines), granodiorite (crosses), and quartz monzonite (solid black). Preplutonic rocks are shown by stippling.

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TABLE 1. Sampling and Remanence Data by Sites

Site	Pluton Phase	Location		Intensity of af Demagnetizing Field,* Oe	Sampling†						Remanent Magnetization				
		Longitude, °W	Latitude, °N		N	a	b	c	d	n'	Resultant Vector R	Declination, deg	Inclination Positive Downward, deg	Precision Parameter k	Radius of 95% Confidence α_{95}, deg
1	Ponder granodiorite	129.00	54.41	100	5		4	1		0					
2	Ponder granodiorite	129.01	54.41	200	5		5			0					
3	Ponder granodiorite	129.14	54.37	150	5		4	1		0					
4	Ponder granodiorite	129.14	54.36	150	5		5			0					
5	Kasiks diorite	129.40	54.30	100	5		2			3	2.947	15.1	73.9	37	13.2
6	Kasiks diorite	129.41	54.29	300	5		2		1	2	1.999	0.3	70.3	701	3.7
7	Kasiks diorite	129.44	54.26	100 (300)	5		1			4	3.935	344.4	72.0	46	10.4
8	Kasiks diorite	129.46	54.25	100 (400)	5		3			2	1.915	329.8	72.1	12	28.9§
9	Quottoon quartz diorite	129.59	54.21	100	5		3			2	1.994	347.2	73.8	175	7.5
10	Quottoon quartz diorite	129.61	54.19	100 (400)	5	1	1			3	2.929	354.5	78.9	28	15.2
11	Quottoon quartz diorite	129.63	54.19	100	5	1	2			2	1.985	344.6	76.2	66	12.2
12	Quottoon quartz diorite	129.66	54.20	100	5	1	3	1		0					
13	Quottoon gneissic diorite	129.74	54.22	100 (400)	5					5	4.982	324.4	81.7	227	4.2
14	Quottoon gneissic diorite	129.75	54.22	100	5	2	1			2	1.852	6.9	60.8	7	38.2§
15	Quottoon gneissic diorite	129.76	54.22	100 (400)	5					5	4.970	30.7	80.8	132	5.5
16	Quottoon gneissic diorite	129.78	54.22	200	5					5	4.939	303.2	83.6	65	7.8
17	Ecstall diorite	129.88	54.23	(100) 400	5					5	4.984	160.8	-76.6	254	3.9
18	Ecstall diorite	129.88	54.23	150	5					5	4.976	209.0	-69.1	166	4.9
19	Ecstall diorite	129.89	54.22	(150) 400	5					5	4.941	221.0	-56.8	68	7.6
20	Ecstall diorite	129.91	54.22	(100) 400	5					5	4.846	203.7	-61.6	26	12.3
21	Ecstall quartz diorite	129.92	54.22	(100) 300	5	1			1	3	2.767	250.3	-13.6	9	27.6§
22	Ecstall quartz diorite	129.94	54.21	150	5					5	4.985	11.8	76.7	267	3.8
23	Ecstall quartz diorite	130.03	54.23	(100) 400	5					5	4.895	49.1	70.3	190	4.6
24	Ecstall quartz diorite	130.07	54.23	300	5					5	4.895	250.3	-51.0	38	10.2
25	Ecstall quartz diorite	130.11	54.24	150	5	4	1			0					

*The rejected af demagnetization treatment is given in parentheses.

†N is the number of cores collected; a, b, c, and d are cores rejected for reasons explained in the text under experimental procedure; and n' is the number of cores used to calculate the site mean statistics.

§ These sites were rejected because of insufficiently homogeneous RM.

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field required to reduce the NRM intensity by one half) exceeded 800 Oe, and the remanence directions were virtually unchanged. Of these 11 sites, 10 were ultimately found to have homogeneous site mean characteristic RM directions. The pilot specimens for the remaining 14 sites gave a type of response that indicated some remanence instability. For these 14 the median destructive field was always less than 80 Oe and in 12 cases less than 40 Oe, which is indicative of unstable viscous RM. With one exception (site 14) all 14 specimens gave increases in remanence intensity associated with the acquisition of anhysteretic RM in high cleaning fields. Significantly, only one of the 14 sites represented proved to have five cores meeting the minimum criteria for acceptability, and only seven sites were ultimately found to have a stable site mean characteristic RM direction.

The following criteria for acceptability were used (Table 1):

a. If specimens had a remanence intensity of less than 5×10^{-7} emu/cm^3, they were rejected to avoid the introduction of errors through instrumental noise on measurement.

b. If the remanence directions of the two specimens from one core diverged by more than 20°, the core was considered inhomogeneously magnetized and was rejected.

c. If after *a* and *b* were applied only one acceptable core remained to represent a site, this was considered insufficient representation for a site, and the core was rejected.

d. If the characteristic RM direction of one core deviated by more than 3 times the circular standard deviation from the mean of the remaining site cores, the core was considered anomalous and was rejected.

The site mean remanence directions were then calculated by giving each core unit weight (Table 1). For three sites (8, 14, and 21) the radius of the cone of 95% confidence (α_{95}) [*Fisher*, 1953] exceeded 20°, indicating an unacceptably large dispersion. These three sites were therefore considered unreliable records of the earth's paleomagnetic field and were rejected from further analysis.

As a result of this screening procedure it was concluded that the samples collected from the granodiorite phase of the Ponder pluton do not retain an identifiable characteristic RM because all sites were rejected.

Statistical Analysis

The acceptable site mean remanence directions were each given unit weight and combined to calculate the phase mean remanence directions (Table 2, Figure 3). All phase mean directions deviate from the earth's present magnetic field direction. This result is evidence for the isolation of a pre-Quaternary remanence. The preservation of antiparallel reversed and normal mean directions in the Ecstall pluton is strong evidence for preservation of a thermal remanent magnetization (TRM) acquired when the rocks last cooled through their Curie temperatures. Examination of the mean directions leads directly to the question of correlation of phases. The angular variance ratio method of *Larochelle* [1969] was employed to test some possible correlations statistically.

The gneissic diorite phase sampled at sites 13–16 gives site mean remanence directions that belong to the same population as those of the Quottoon quartz diorite sites at the 95% confidence level (test 1, Table 3) and to a different population from those of the Ecstall diorite sites (test 3). This result is reinforced by the observation that the gneissic diorite and the Quottoon quartz diorite phases retain a normal polarity, whereas the Ecstall diorite phase retains a reversed polarity. The consistent remanence directions within the gneissic diorite phase up to 3 km away from its contact with the Quottoon quartz diorite suggest that the gneissic diorite is not a contact-metamorphosed phase of the Ecstall pluton. Therefore it is thought that the gneissic diorite phase belongs to the Quottoon pluton and that it is essentially comagmatic and coeval with the quartz diorite phase. Accordingly, the sites from both phases were combined to obtain the mean remanence direction for the Quottoon pluton (Table 3, Figure 3). The close grouping of the site mean remanence directions of all normal polarity further indicates that all sites cooled through their Curie temperatures during one polarity interval. *Cox* [1968, 1969] and *Kono* [1973] have shown that the average duration of a magnetic interval during the last 80 m.y. is $\simeq$¼ m.y., that the maximum duration is $\simeq$1 m.y., and that the total times of normal and reversed polarities are about equal. This finding is important because it indicates that the sampled portion of the Quottoon pluton cooled quickly within one normal polarity interval, or in less than $\simeq$½ m.y., in all probability. Similarly, the Kasiks pluton must have cooled within one normal polarity interval but probably not the same interval as that for the Quottoon pluton. This correlation between intrusive phase and remanence direction also indicates that these rocks retain a primary Eocene TRM.

Test 2 (Table 2) indicates that the diorite and quartz diorite phases of the Ecstall pluton are correlative after the polarity of some sites is reversed. Therefore it is probable that the two phases are comagmatic, that they have experienced the same

TABLE 2. Mean Remanence Directions for Groups of Sites

Group	Sites	Number of Sites	Mean Remanence Direction: Resultant Vector R	Declination, deg	Inclination Positive Downward, deg	Precision Parameter k	Radius of 95% Confidence α_{95}, deg
By petrologic phase							
Kasiks diorite	5 to 7	3	2.9926	359.4	72.5	272	4.9
Quottoon quartz diorite	9 to 11	3	2.9976	348.3	76.3	838	2.8
Quottoon gneissic diorite	13, 15, 16	3	2.9886	343.4	83.6	175	6.1
Ecstall diorite	17 to 20	4	3.9307	205.1	-67.2	43	10.7
Ecstall quartz diorite	22 to 24	3*	2.9119	53.6	67.6	23	17.0
By pluton							
Kasiks	5 to 7	3	2.9926	359.4	72.5	272	4.9
Quottoon	9 to 11, 13, 15, 16	6	5.9739	346.7	80.0	191	4.1
Ecstall	17 to 20, 22 to 24	7†	6.8126	37.1	68.0	32	9.4
By age							
Eocene	5 to 7, 9 to 11, 13, 15, 16	9	8.9467	352.6	77.6	150	3.8
Upper Cretaceous	17 to 20, 22 to 24	7†	6.8126	37.1	68.0	32	9.4

*Remanence direction of site 24 rotated by 180°.
†Remanence directions of sites 17 to 20 and 24 rotated by 180°.

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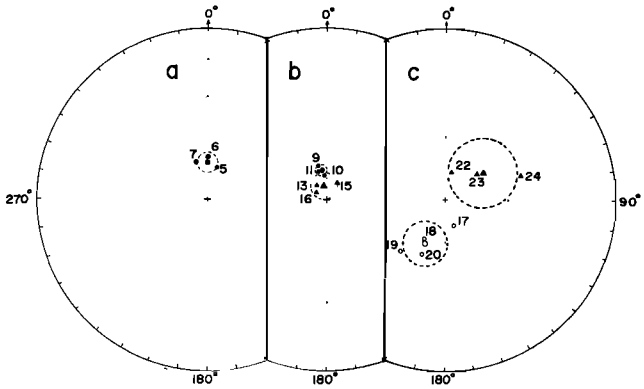


Fig. 3. Schmidt equal area stereographic projections showing the site mean remanence directions for (*a*) the diorite phase of the Kasiks pluton, (*b*) the quartz diorite (circles) and gneissic diorite (triangles) phases of the Quottoon pluton, and (*c*) the diorite (circles) and quartz diorite (triangles) phases of the Ecstall pluton. Downward normal (positive) and upward reversed (negative) vectors are shown as solid and open symbols, respectively. The polarity of site 24 has been 'reversed.' The phase mean direction is shown by a larger symbol surrounded by its cone of 95% confidence.

tectonic history, and that their site mean directions can be combined to obtain a plutonic mean direction to compare with other Upper Cretaceous results. Because of their mixed polarity it is clear that all sites are not strictly coeval. Because sites 22 and 23 are closest to the core of the pluton, it is probable that only one polarity change of the earth's paleomagnetic field, or two magnetic intervals, are represented. It therefore appears that all the sites cooled through their Curie temperatures in less than $\simeq 1$ m.y. during Upper Cretaceous time.

Test 4 (Table 2) compares the direction populations of the Kasiks and Quottoon plutons. The test gives a barely negative result despite the very small radii of the cones of 95% confidence (α_{95}) of the two populations. This result suggests that although these two plutons are not comagmatic or coeval, they are very similar in tectonic evolution and in age, as is suggested by their Eocene K-Ar ages. Therefore the site mean directions for the two plutons have been combined to give a mean remanence direction that can be compared with other Eocene data.

Pattern of Remanence Directions

The earth's magnetic field reversed polarity some 120 times in the 30-m.y. time interval between the K-Ar ages recorded for the 43-m.y.-old Kasiks pluton and the 73-m.y.-old Ecstall pluton. If the K-Ar ages record the age of uplift, unroofing, and cessation of argon diffusion, then the isotherms associated with the uplift must have migrated downward and eastward to give the pattern of K-Ar age zones found in the Coast plutonic complex. Similarly, if the K-Ar ages record regional thermal metamorphism associated with the North American plate boundary, then the isotherms must also have migrated downward and eastward. Both these hypotheses would be expected to give the same sort of pattern of site mean remanence directions for the E-W Skeena River section. For both, the directions would be independent of petrologic phase or plutonic boundaries. Also, on progressing easterly along the section the sites should have an alternating pattern of polarities because each has an equal chance of acquiring a normal or reversed polarity. The site directions do not have such a pattern but rather form coherent clusters related to phase and plutonic boundaries. Thus the directions record the emplacement and rapid cooling of the plutons. This result supports the hypothesis that the K-Ar ages date the time of emplacement and cooling.

Pole Positions

The Eocene pole calculated from the mean remanence direction obtained from the combined Kasiks and Quottoon sites and the Upper Cretaceous pole calculated from the Ecstall sites deserve further comment. Listed in Table 4 are pole positions from North American rock units of Upper Cretaceous to Oligocene age that are based on remanence data from several samples from at least five separate sites.

The Quottoon-Kasiks pole (pole 10, Figure 4) is concordant with other lower Tertiary pole positions from the stable cratonic region of North America. This group includes the Cape Dyer volcanics (pole 7), the Green River red beds (pole 9), the Spanish Peaks dikes pole (pole 11), and the west central United States volcanics (pole 15). It is concluded that the Quottoon and Kasiks plutons acquired their remanence during the Eocene, probably as a primary TRM, and that they have not since been rotated significantly or translated. This is the same conclusion as that reached for the cordilleran Hope pluton (pole 13 [*Symons*, 1973*c*]). All other lower Tertiary pole positions for cordilleran units are discordant. The discordance was likely caused by regional tectonic deformation in the case of the Siletz River volcanics (pole 8 [*Wise*, 1963; *Irving*, 1964, p. 249]) and of the East Sooke gabbro (pole 12 [*Symons*, 1973*b*]) or by incomplete averaging of secular varia-

TABLE 3. Angular Variance Ratio Analysis Between Units

Test	Units Compared	Number of Sites B	Variance Ratio δ_b^2/δ_w^2	Theoretical Statistic $F_{0.05}$	Correlation Test Result
1	Quottoon quartz diorite and Quottoon gneissic diorite	6	3.57	4.46	Positive
2	Ecstall diorite and Ecstall quartz diorite	7	0.98	4.10	Positive
3	Quottoon gneissic diorite and Ecstall diorite	7	5.54	4.10	Negative
4	Kasiks pluton and Quottoon pluton	9	4.16	3.74	Negative

The variance ratio δ_b^2/δ_w^2 is the between-unit over the within-unit angular variances in (radians)^2. The theoretical statistic if $F_{0.05} = F_{2,2(B-2),0.05}$, thereby setting the test at the 95% confidence level.

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TABLE 4. Paleomagnetic Pole Positions From North American Rock Units of Upper Cretaceous and Lower Tertiary Age

Pole	Rock Unit	Locality	Age, m.y.	Number of Sites (Samples)	Longitude, °W	Latitude, °N	Axes of Oval of 95% Confidence, deg δ_p	δ_m	Reference
			Upper Cretaceous						
1	Alkalic complexes	Arkansas	97 ± 10	19	170	70	9	13	*Scharon and Hsu* [1969]
2	Howe Sound plutons	SW British Columbia	94 ± 6	17	231	83	10	13	*Symons* [1973a]
3	Sierra Nevada pluton	California	86 ± 4	14	165	69	10	10	*Gromme and Merrill* [1965]
4	Elkhorn Mountain volcanics	SW Montana	80	13	171	69	7	10	*Hanna* [1967]
5	Boulder batholith	Montana	75 ± 7	(15)	193	76	6	7	*Hanna* [1967]
6	Ecstall pluton	NW British Columbia	75 ± 12	7	47	68	13	16	This study
			Paleocene-Eocene						
7	Cape Dyer lavas	Baffin Island, N. W. T.	58 ± 2	5	55	83	12	12	*Deutsch et al.* [1971]
8	Siletz River volcanics	Oregon	53	8	49	37	7	10	*Cox* [1957]
9	Green River red beds	Wyoming	50	7	168	85	8	9	*Torreson et al.* [1949]
10	Quottoon and Kasiks plutons	NW British Columbia	47 ± 6	9	144	76	7	7	This study
11	Spanish Peaks dikes	Colorado	40	6	149	81	16	20	*Larson and Stranway* [1969]
12	East Sooke gabbro	Vancouver Island, B. C.	39	29	209	70	5	7	*Symons* [1973b]
			Oligocene						
13	Hope granitic pluton	SW British Columbia	38 ± 3	5	152	88	5	6	*Symons* [1973a]
14	Mount Washington	Vancouver Island, B. C.	37 ± 5	5	157	64	10	10	*Symons* [1971b]
15	Oligocene volcanics	West Central U.S.A.	30 ± 8	67	214	78	9	11	*Gromme and McKee* [1971]
16	Marys Peak sill	Oregon	30 ± 2	24	164	63	6	10	*Clark* [1969], *Hicken et al.* [1972]

tion of the earth's magnetic field to give a virtual geomagnetic pole in the case of the Mount Washington dacites (pole 14 [*Symons*, 1971*b*]) and of the Marys Peak sill (pole 16 [*Clark*, 1969]) which cooled of the order of 4000 ± 2000 years.

Four Upper Cretaceous poles from the North American craton or from the relatively stable and slightly disturbed eastern and central regions of the cordillera (poles 1, 3, 4, and 5; Table 4 and Figure 4) form a consistent cluster. The two poles from the western side of the cordillera are divergent. The discordance of the pole of the Howe Sound plutons is not statistically significant and has been explained in terms of a few degrees of SW tilt of the plutons [*Symons*, 1973*a*]. The discordance of the Ecstall pole (6) is significant. Its position is not compatible with any portion of the North American Phanerozoic polar wander path. This incompatibility is certainly the result of structural deformation. To begin with, the discordance provides a sort of 'fold test' in that the mixed polarity remanence of the Ecstall pluton must be stable and predate the time of deformation.

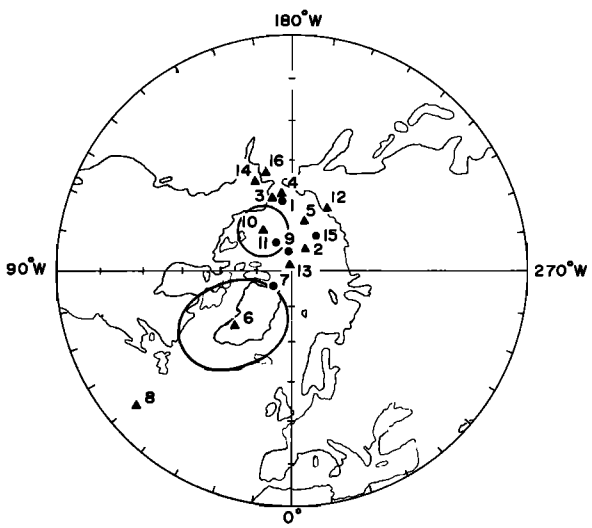


Fig. 4. Polar projection of the northern hemisphere above latitude 30°N. Poles from North American cratonic and cordilleran rock units are distinguished by circles and triangles, respectively. The pole numbers correspond to the list in Table 4. The ovals of 95% confidence have been provided for the two new poles reported in this study.

On the basis of paleomagnetic data from the Stevens Pass granites, *Beck and Noson* [1972] suggested that the Cascade Mountains have been translated northward by some 30° of arc relative to the North American craton between the mid-Upper Cretaceous and latest Eocene times [*Beske et al.*, 1973]. If the Cascade Mountains have been subjected to essentially the same geotectonic history as the Coast plutonic complex, then this hypothesis is incompatible with the concordant pole positions found for the Eocene Quottoon-Kasiks pluton (pole 10), the Howe Sound plutons (pole 2), and the Jurassic island intrusions from Vancouver Island in the insular fold belt [*Symons*, 1971*a*], which all come from units in the extreme western cordillera. In the case of the Ecstall pluton, translation would have to be primarily from west to east, a conclusion incompatible with the geologic evidence.

The Ecstall pluton pole (6) can be brought into agreement with the known Upper Cretaceous position (poles 1, 3, 4, and 5) by correcting for tectonic rotation. A 50° clockwise twist rotation of the pluton about an adjacent vertical axis would give agreement. However, the major structural, stratigraphic, and petrologic trends are rather uniformly NW-SE throughout this region of the Coast plutonic complex. This conflict suggests that any rotation must be dominantly about a horizontal axis. A 20°W tilt of the pluton about a horizontal N-S axis would also produce agreement. The Ecstall pluton is part of a tectonic block or panel that is bounded by steeply dipping faults on the SW side along Grenville Channel [*Roddick*, 1970] and Malacca Passage and on the northern part of the NE side along Work Channel [*Hutchison*, 1967]. Starting with an originally horizontal plane a net tilt of 20°W about a N-S horizontal axis would be the resultant of (1) a 15°NW tilt or rotation of the panel about a horizontal NE-SW axis (i.e., an axis perpendicular to the plane of the panel), together with (2) a 15°SW tilt or rotation about a horizontal NW-SE axis (i.e., an axis in the plane of the panel). Evidence for tilt 1 comes from the distribution of phases in the Ecstall pluton. This pluton is a para-autochthonous, steep-walled, tadpole-shaped pluton in which the 'head' of the tadpole is more homogeneous, massive, and acidic than the 'tail' [*Hutchison*, 1967]. This finding suggests under a diapiric intrusive model that the head was formed at a higher level than the tail. Because the head is toward the NW end of the panel, this sug-

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gestion is consistent with a low-angle NW tilt. Evidence for tilt 2 comes from the contact and foliation plane attitudes that dip vertically to steeply east (≃60°) [*Roddick*, 1970]. Such dips are consistent with a ≃15°SW tilt. Unfortunately, there are no overlying stratiform rocks to confirm these rotations, although additional paleomagnetic work on plutons occupying tectonic panels to the SW of this panel would be very revealing.

CONCLUSIONS

1. The granodiorite phase of the Ponder pluton has not retained a stable characteristic RM.

2. The gneissic diorite phase outcropping between the Quottoon and Ecstall plutons is part of the Quottoon pluton.

3. The Quottoon and Kasiks plutons retain stable well-grouped normal characteristic RM directions. Their pole positions individually and combined are consistent with other Eocene or lower Tertiary pole positions from tectonically undisturbed rock units of the North American craton.

4. The Kasiks and Quottoon plutons cooled within one normal polarity interval, i.e., within a time span of less than ≃½ m.y. This rather rapid rate of cooling strongly supports the hypothesis that the remanence and K-Ar dates record the cooling upon emplacement of the plutons. Cooling by progressive unroofing of the plutons from west to east should be reflected in the easterly migration of isotherms giving site mean remanence directions of mixed normal and reversed polarities that are independent of petrologic phases and plutonic boundaries. The same directional pattern would be expected if the cooling resulted from the easterly migration of some regional thermal metamorphic event associated with the western margin of the North American plate. Such a pattern was not found; the directions are distinctly related to phase and plutonic boundaries.

5. It follows from point 4 that the Kasiks and Quottoon rocks retain a primary TRM of Eocene age. Although it is possible, it seems improbable that both plutons would cool quickly through their Curie temperatures to freeze their magnetic domains and thereafter cool slowly through the somewhat lower temperature range in which the K-Ar mineral chemical systems are closed.

6. The Ecstall pluton retains a stable characteristic RM of mixed polarity that is also likely a primary TRM. The sampled portion of the Ecstall pluton probably cooled within a time span of less than ≃1 m.y. during the Upper Cretaceous, based on the pattern of site polarities and the preservation of a predeformational remanence.

7. The Ecstall pluton is part of a tectonic block or panel bounded by NW-trending faults. The panel has a net tilt of ≃20°W.

8. If it is accepted that the K-Ar ages record the date of emplacement, then any evolutionary model for the Coast plutonic complex must explain the three age zones. In general terms this requires a prolonged pulse of intrusive activity along the western margin of the North American plate from 84 to 140 m.y. ago during which the plutons in the western age zone were emplaced. The center of igneous activity then moved inland to the east giving an intrusive pulse between 64 and 80 m.y. ago to form the central zone and again further inland between 40 and 50 m.y. to form the eastern zone. More paleomagnetic work on the central and western zones is needed and planned to substantiate this hypothesis.

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